SOLUTION REQUIREMENTS

Project Name: EduGenie

Technology: Google Gemini Generative AI

Project Type: AI-Powered Learning Assistant

1. INTRODUCTION

EduGenie is an AI-powered learning assistant designed to help students understand and revise educational content. The system uses Google Gemini Generative AI to process user-provided educational content and generate useful, concise and understandable responses.

The solution requirements define the functional and non-functional requirements needed for the EduGenie system.

2. FUNCTIONAL REQUIREMENTS

2.1 User Input

The system shall allow students to enter or provide educational content.

The input may include:

- Study material.
- Educational text.
- Learning-related questions or instructions.

2.2 Input Processing

The system shall receive and process the user's input before sending it to the AI service.

The system should:

- Accept valid user input.
- Process the input correctly.
- Prepare an appropriate prompt for the AI model.

2.3 Google Gemini Integration

The system shall integrate with Google Gemini Generative AI.

The integration shall allow EduGenie to:

- Send the user's input to Gemini.
- Receive the generated response.
- Process the returned response for display.

2.4 AI-Generated Response

The system shall generate useful learning content based on the user's input.

The response may include:

- Concise summaries.
- Simple explanations.
- Important information.
- Learning-related answers.

2.5 Result Display

The system shall display the generated response to the student in a clear and readable format.

The student should be able to easily read and understand the generated content.

2.6 Error Handling

The system should handle common errors such as:

- Empty user input.
- Invalid requests.
- Internet connectivity problems.
- AI service/API errors.
- Unexpected responses.

3. NON-FUNCTIONAL REQUIREMENTS

3.1 Usability

The system should have a simple and student-friendly interface so that users can interact with the application easily.

3.2 Performance

The system should process user requests efficiently and return the generated response within a reasonable time, depending on network and AI service availability.

3.3 Reliability

The application should handle user requests consistently and provide appropriate feedback when an error occurs.

3.4 Security

Sensitive configuration information such as API keys should not be exposed directly in the application's user interface or public source code.

3.5 Maintainability

The project should use a clear code structure so that future developers can modify existing features and add new functionality easily.

3.6 Scalability

The system should be designed so that additional AI-powered learning features can be added in future versions.

4. HARDWARE REQUIREMENTS

Minimum requirements:

- Computer or laptop.
- Internet connectivity.
- Modern processor.
- Minimum 4 GB RAM.
- Sufficient storage for the application and project files.

5. SOFTWARE REQUIREMENTS

The project may require:

- Python.
- Required Python packages.
- Google Gemini API / Generative AI service.
- A suitable code editor or IDE.
- Modern web browser, if the application uses a web interface.

• Internet connection.

6. USER REQUIREMENTS

The user should be able to:

• Open the EduGenie application.

• Enter educational content.

• Submit the content for AI processing.

• Receive the generated response.

• Read and use the response for learning and revision.

7. FUTURE REQUIREMENTS

Future versions of EduGenie can include:

• AI-powered question answering.

• Quiz generation.

• Flashcard generation.

• Personalized study plans.

• Topic-wise learning.

• Progress tracking.

• Multiple learning modes.

8. CONCLUSION

The solution requirements provide the basic technical and user requirements for developing EduGenie. The main requirement is to provide a simple interface through which students can submit educational content and receive useful AI-generated learning assistance using Google Gemini.